디지털논리회로설계

Digital Logic Circuit Design

(VHDL Code Result Report)

name : 양유경

major : 로봇공학과 (Robotics)

student ID : 2025019425

Course Number : 23278

Date : October 13, 2026

1. VHDL code

-1. and_or_out

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
ENTITY and_or_out IS
 PORT (
  a, b, c, d : in std_logic;
  out1, out2 : out std_logic
 );
END and_or_out;

 --architecture
architecture bhv of and_or_out is
begin
  out1 <= a and b and c and d;
  out2 <= a or b or c or d;
end architecture bhv;
```

-2. multiplexer_8_1

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
ENTITY multiplexer_8_1 IS
 PORT (
 a, b, c, d, e, f, g, h : in std_logic;
 ch : in std_logic_vector(2 downto 0);
 data : out std_logic
 );
END multiplexer_8_1;

--arcitecture
ARCHITECTURE bhv OF multiplexer_8_1 IS
BEGIN
  process(a, b, c, d, e, f, g, h, ch)
 begin
 case ch is
 when "000" => data <= a;
 when "001" => data <= b;
```

```
 when "010" => data <= c;
 when "011" => data <= d;
 when "100" => data <= e;
 when "101" => data <= f;
 when "110" => data <= g;
 when others => data <= '0';
 end case;
 end process;
END bhv;
```

-3. digital_letters

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

--entity
entity digital_letters is
 port (
 en : in std_logic;
 dat : in std_logic_vector(2 downto 0);
 display : out std_logic_vector(7 downto 0)
 );
end digital_letters;


--architecture
architecture bhv of digital_letters is
begin
 process(en, dat)
 begin
 if en = '0' then
 display <= "ZZZZZZZZ";
 else
 case dat is
 when "000" => display <= "11101110"; -- A
 when "001" => display <= "00111110"; -- b
 when "010" => display <= "10011100"; -- C
 when "011" => display <= "01111010"; -- d
 when "100" => display <= "10001110"; -- F
 when "101" => display <= "11110110"; -- g
 when others => display <= "00000000";
 end case;
 end if;
 end process;
end architecture bhv;
```

-4. voting_with_25

```vhdl
LIBRARY ieee;
USE ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;


entity voting_with_25 is
 port (
 vt : in std_logic_vector(25 downto 1);
 result : out std_logic
 );
end entity voting_with_25;

--architecture
architecture bhv of voting_with_25 is
begin
 process(vt)
 variable sum : integer range 0 to 25;
 begin
 sum := 0;
 for i in 1 to 25 loop
 if (vt(i) = '1') then
 sum := sum + 1;
 if sum >= 13 then
 result <= '1';
 else
 result <= '0';
 end if;
 end if;
 end loop;
 end process;
end architecture bhv;
```

-5. vote_display

```vhdl
library ieee;
use ieee.std_logic_1164.all;

--entity
entity vote_display is
 port (
 en : in std_logic;
 vt : in std_logic_vector(15 downto 1);
 display : out std_logic_vector(7 downto 0)
 );
end entity vote_display;

--architecture
architecture bhv of vote_display is
begin
 process(en, vt)
 variable sum : integer range 0 to 15;
 begin
```

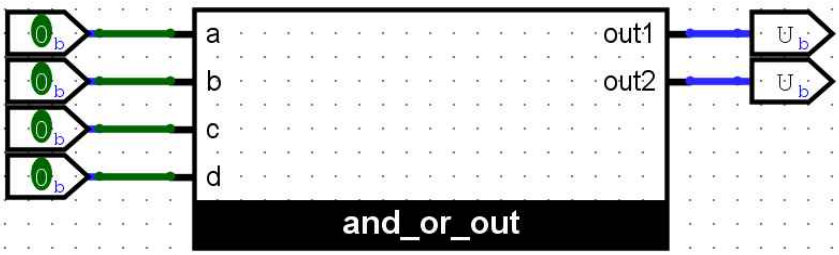
```
 sum := 0;
 for i in 1 to 15 loop
 if vt(i) = '1' then
 sum := sum + 1;
 end if;
 end loop;

 if en = '1' then
 case sum is
 when 0 => display <= "11111101"; -- 0
 when 1 => display <= "01100001"; -- 1
 when 2 => display <= "11011011"; -- 2
 when 3 => display <= "11110011"; -- 3
 when 4 => display <= "01100111"; -- 4
 when 5 => display <= "10110111"; -- 5
 when 6 => display <= "10111111"; -- 6
 when 7 => display <= "11100001"; -- 7
 when 8 => display <= "11111111"; -- 8
 when 9 => display <= "11110111"; -- 9
 when 10 => display <= "11101111"; -- A
 when 11 => display <= "00111111"; -- b (B)
 when 12 => display <= "10011101"; -- C
 when 13 => display <= "01111011"; -- d (D)
 when 14 => display <= "10011111"; -- E
 when 15 => display <= "10001111"; -- F
 when others => display <= "00000000";
 end case;
 else

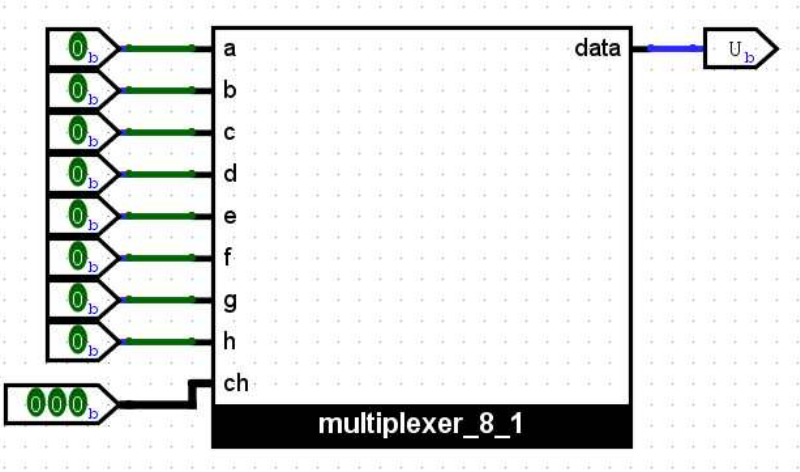
 display <= "00000000";
 end if;
 end process;
end architecture bhv;
```

2. Circuit

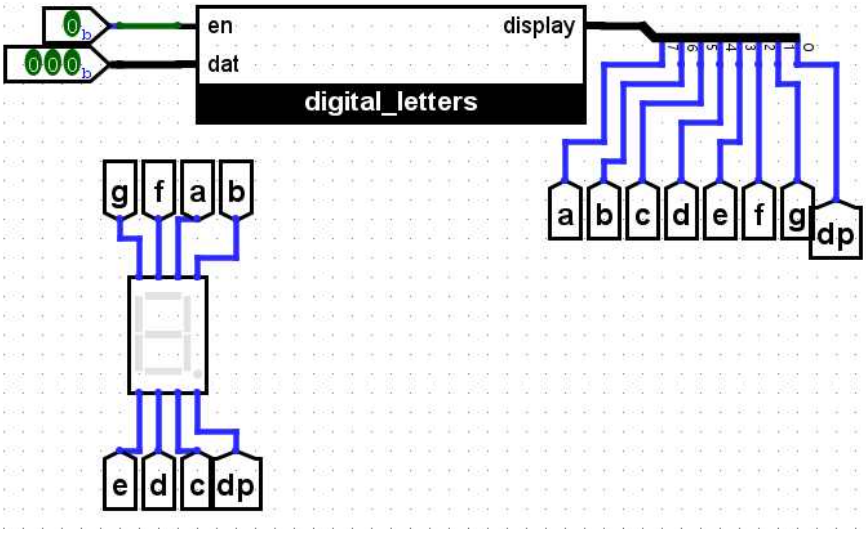
-1. and_or_out



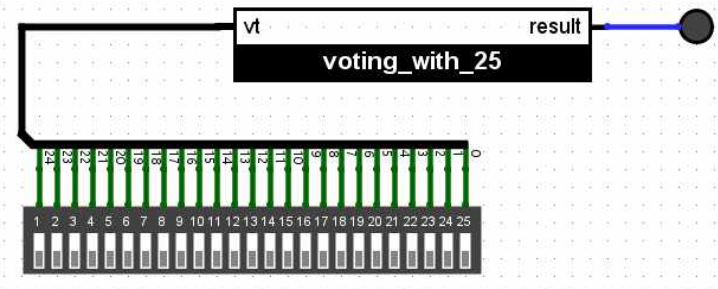
-2. multiplexer_8_1



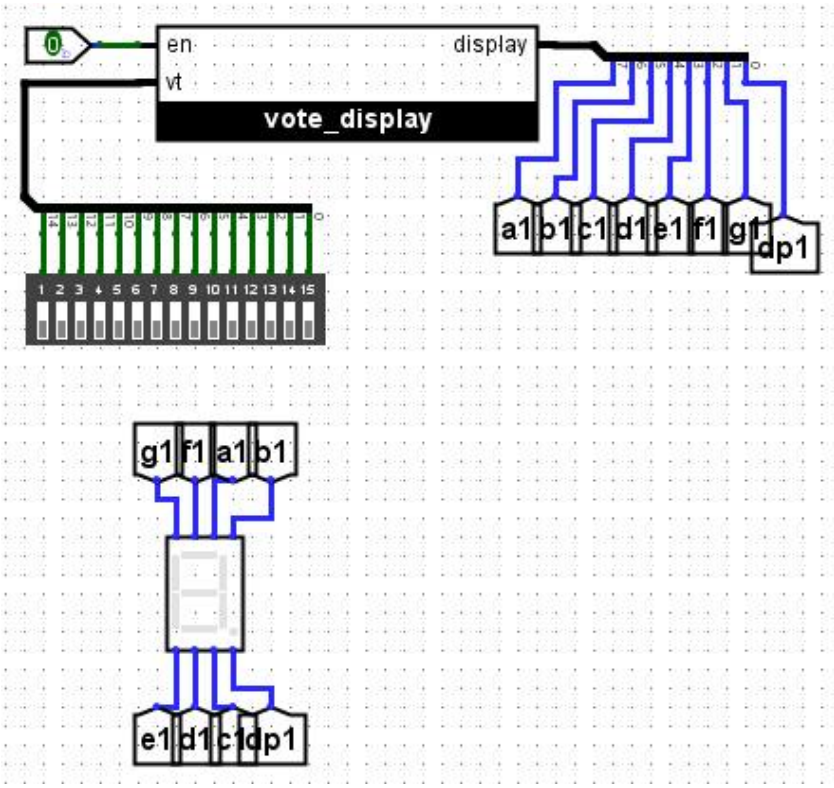
-3. digital_letters



-4. voting_with_25

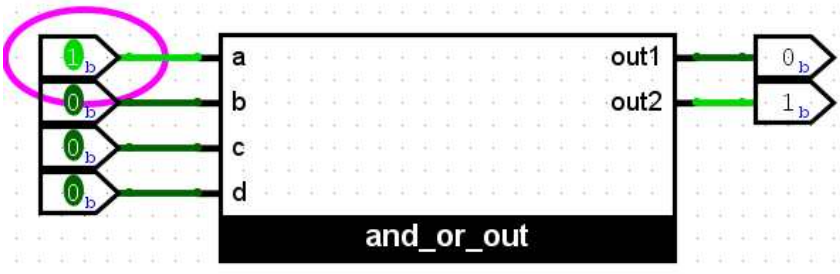


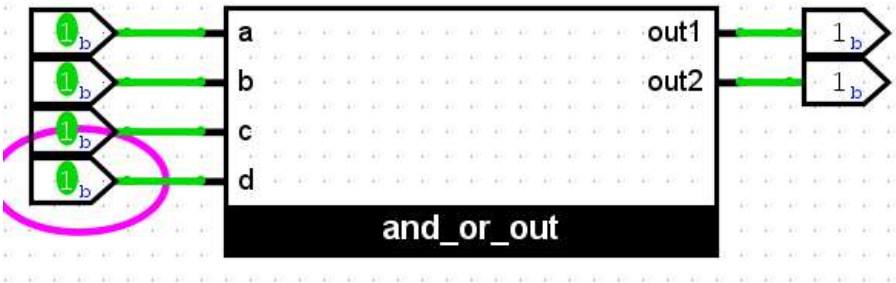
-5. vote_display



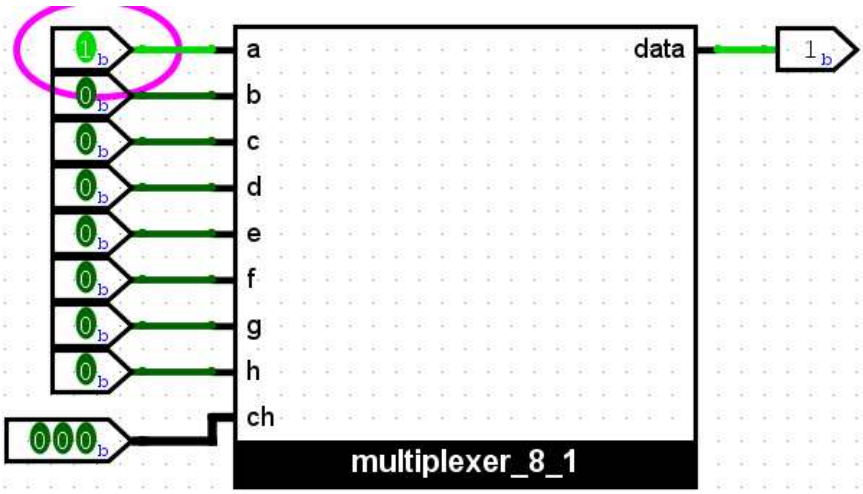
3. Simulation screenshot

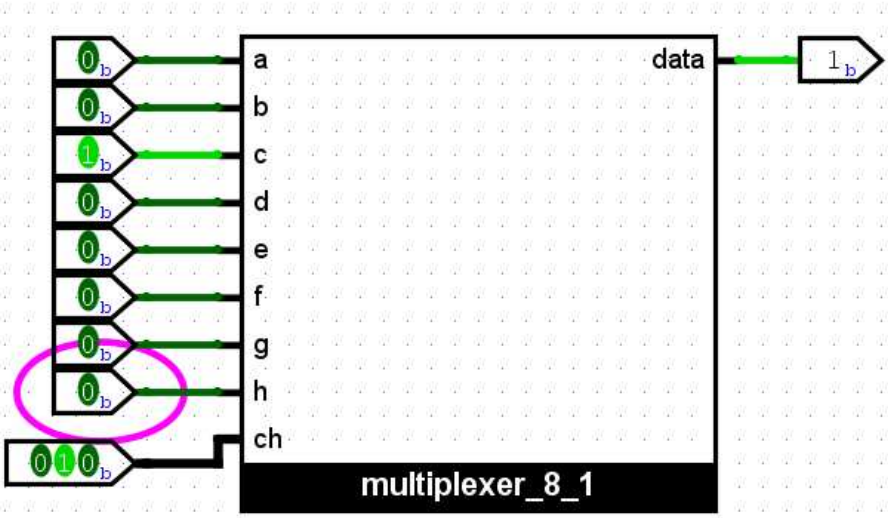
-1. and_or_out



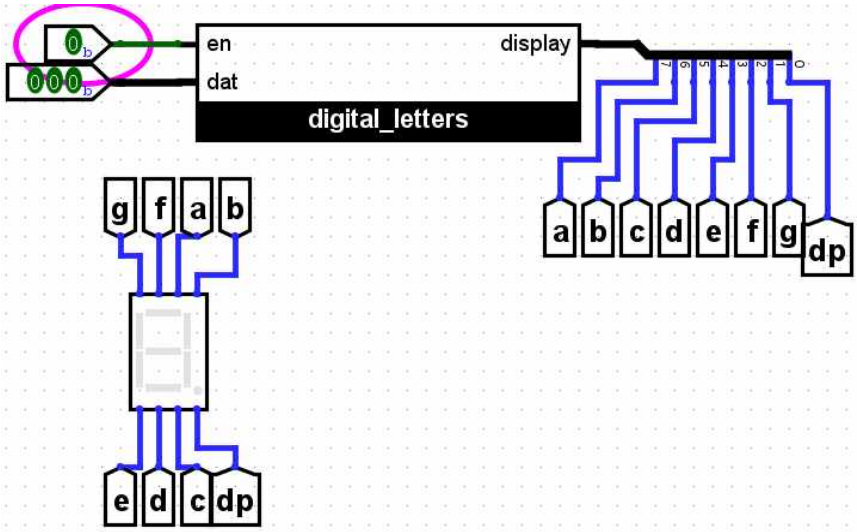


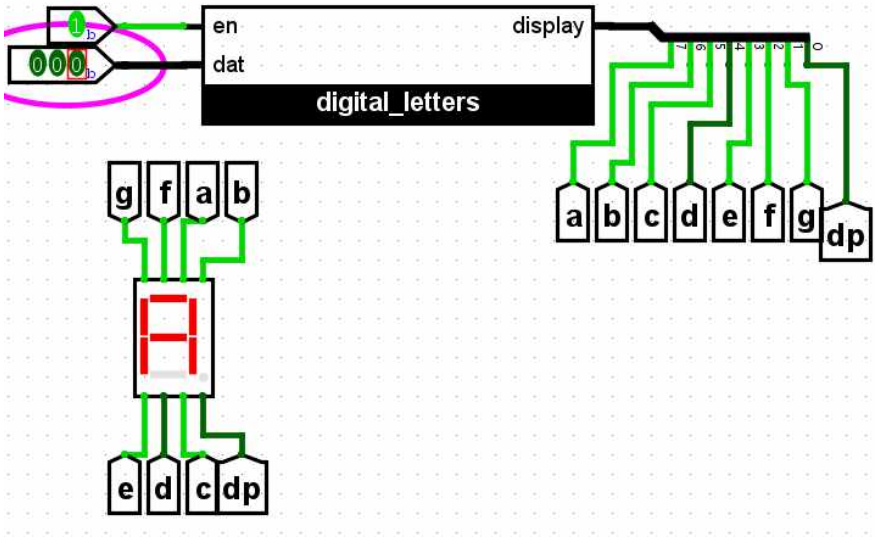
-2. multiplexer_8_1

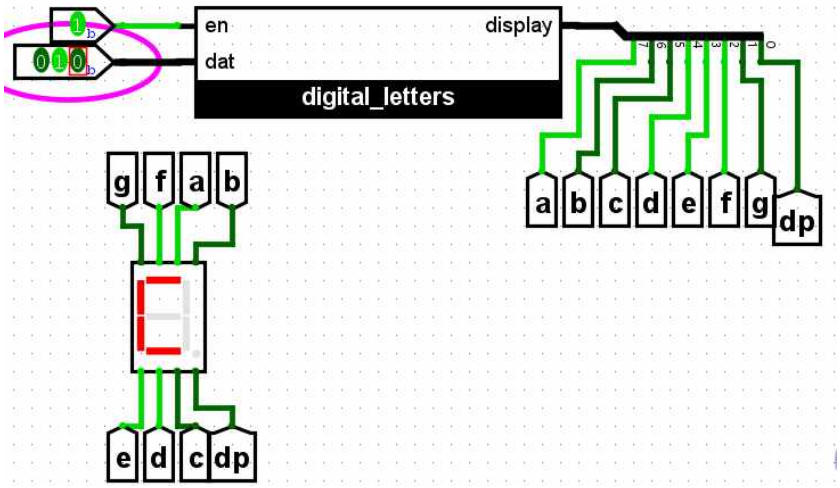


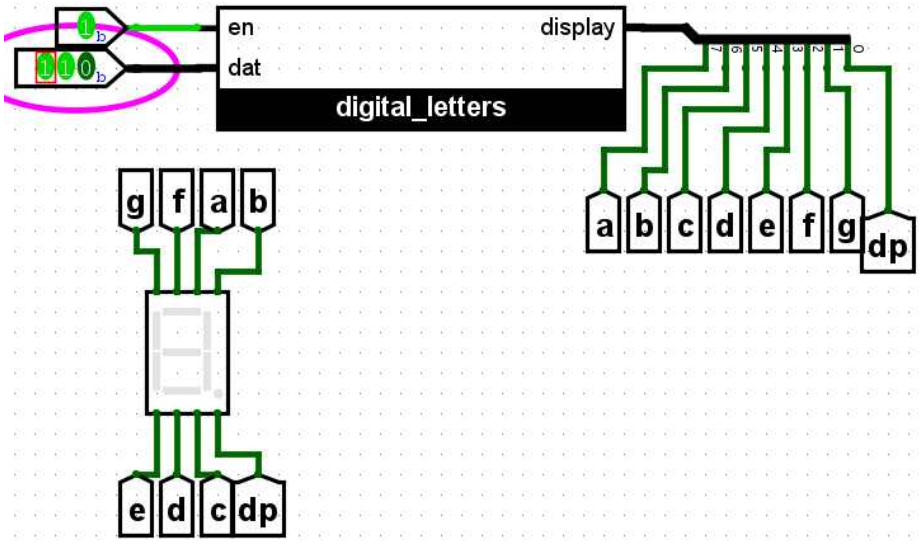


-3. digital_letters

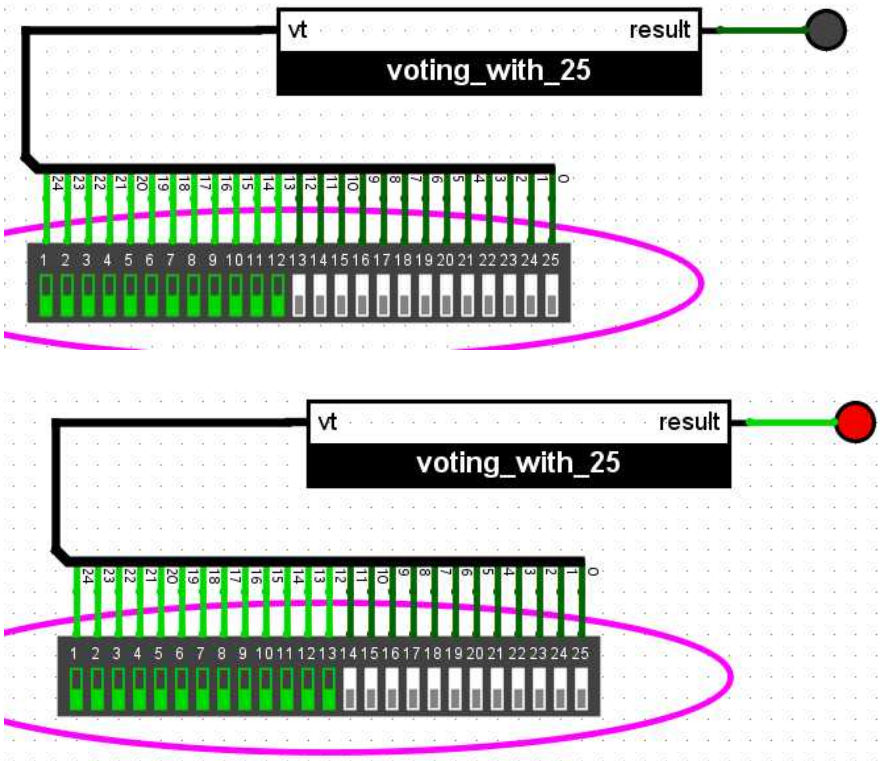

en
dat
display
digital_letters
g f a b
e d c dp
a b c d e f g dp


en
dat
display
digital_letters
g f a b
e d c dp
a b c d e f g dp

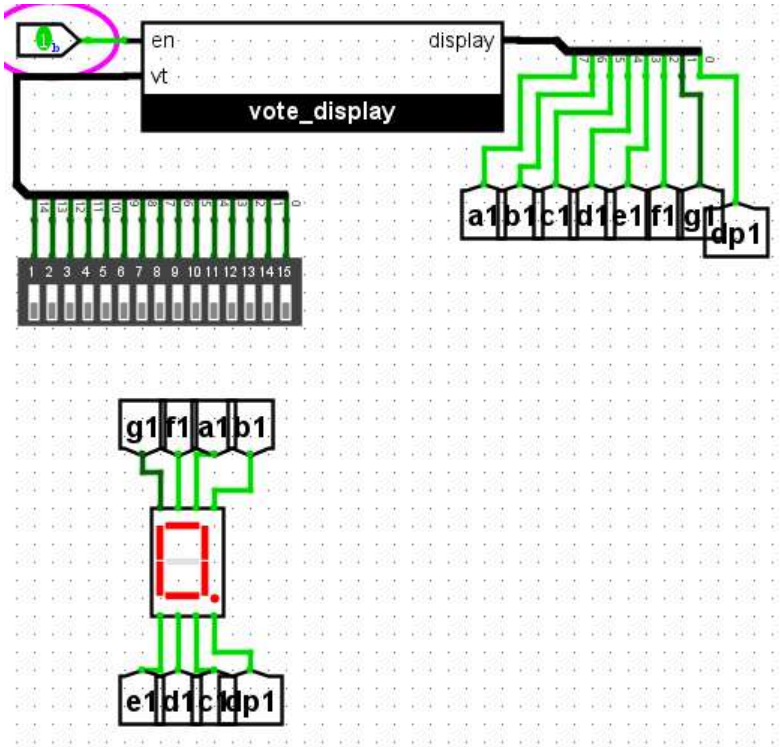

en
dat
display
digital_letters
g f a b
e d c dp
a b c d e f g dp

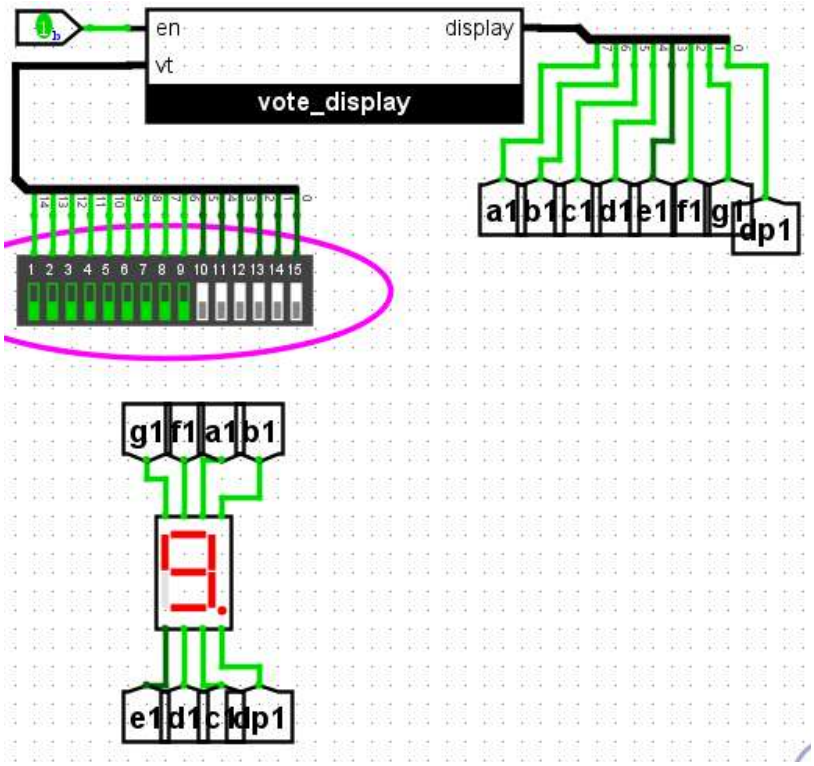


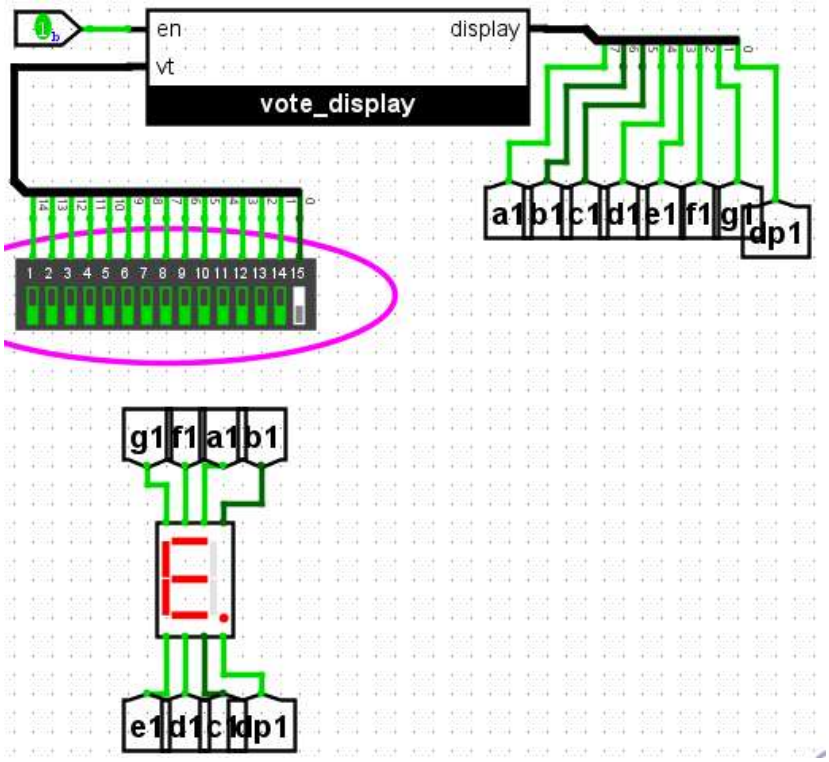
-4. voting_with_25

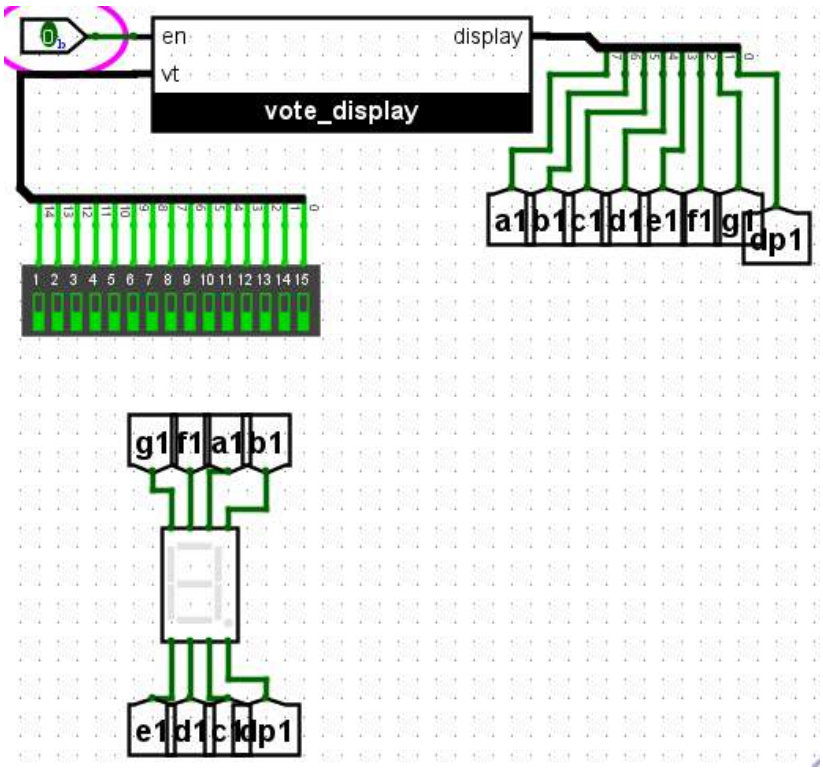


-5. vote_display


en
vt
display
vote_display
a1 b1 c1 d1 e1 f1 g1 dp1
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
g1 f1 a1 b1
e1 d1 c1 dp1


en
vt
display
vote_display
a1 b1 c1 d1 e1 f1 g1 dp1
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
g1 f1 a1 b1
e1 d1 c1 dp1


en
vt
display
vote_display
a1 b1 c1 d1 e1 f1 g1 dp1
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
g1 f1 a1 b1
e1 d1 c1 dp1


en
vt
display
vote_display
a1 b1 c1 d1 e1 f1 g1 dp1
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
g1 f1 a1 b1
e1 d1 c1 dp1

4. Result explanation

The simulation results show that all five VHDL circuits operate according to their designed functions. The AND/OR circuit performs logical operations, and the 8-to-1 multiplexer selects an output based on the selection inputs. The digital letters circuit displays different letters according to the input data and enable signal. The 25-input voting circuit outputs 1 when at least 13 votes are 1, while the vote display circuit counts the active inputs and displays the result in hexadecimal format. These results confirm the basic functions of each circuit.